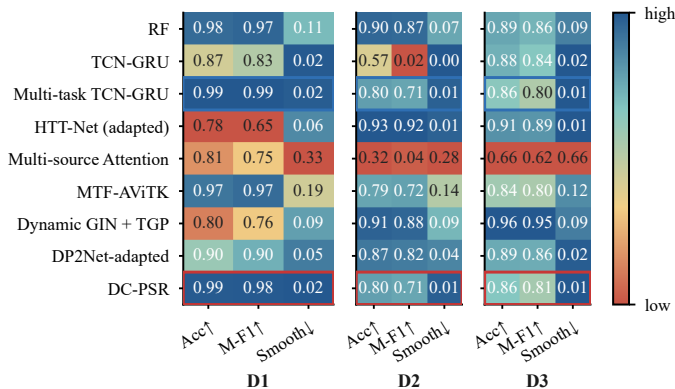
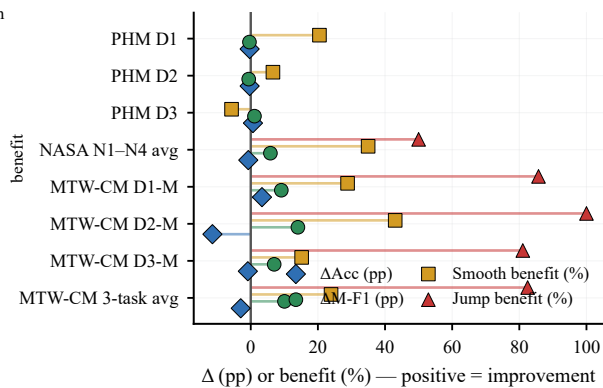
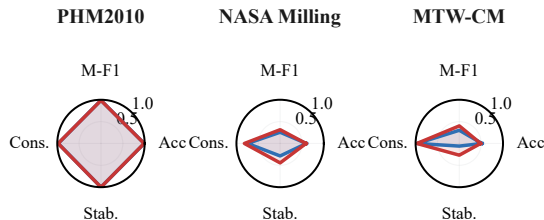




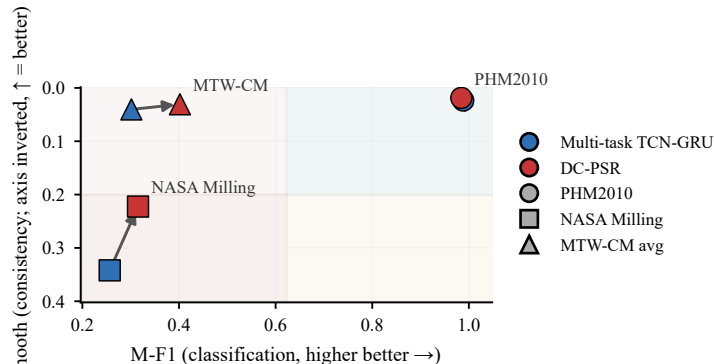
(a) Within-task robustness, PHM2010 D1/D2/D3 $\times$ 9 methods (color = within-task benefit; numbers = raw values)



(b) Multi-task TCN-GRU $\rightarrow$ DC-PSR paired gain, PHM/NASA/MTW-CM (Smooth/Jump = benefit = backbone - DC-PSR; positive = improvement)



(c) Dataset-profile cards: Multi-task TCN-GRU (blue) vs DC-PSR (red) only



(d) Classification-consistency balance: 6 real points, B11 $\rightarrow$ B12 connector per dataset